SCHOOL OF
INTERNATIONAL & INTERDISCIPLINARY
ENGINEERING PROGRAMS
King Mongkut's Institute of Technology Ladkrabang

Week 4: Linked Lists

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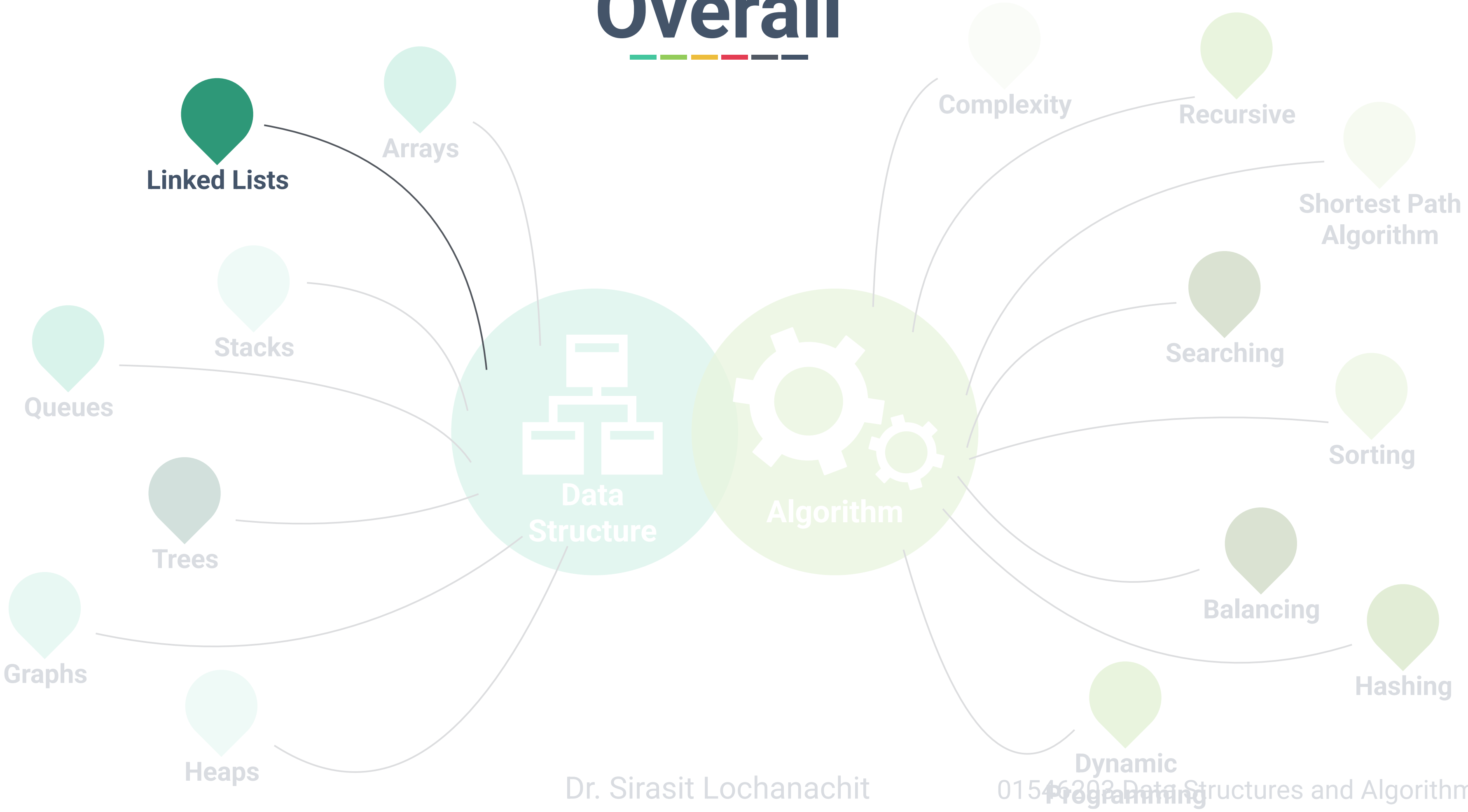
Overall



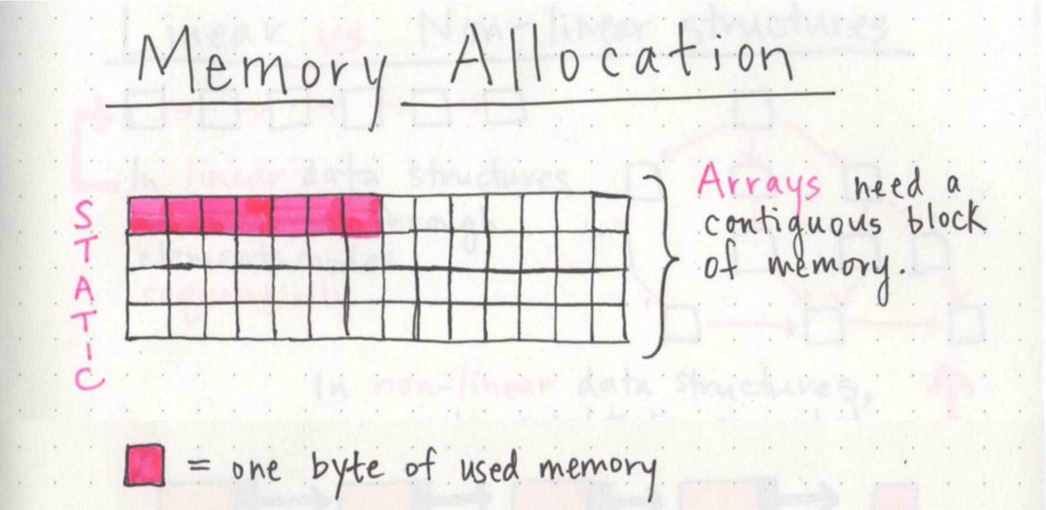
Today's Outline

1. What is a Linked List?
2. Singly Linked Lists
   - Traversing
   - Insert a node
   - Delete a node

Previously

Disadvantages of array:

- Length of array has to be **pre-allocated and fixed**, empty space wasted and unable to grow.
- Adding or removing elements between values in the array is expensive - $O(n)$



Linked Lists

To avoid these limitations, an alternative to array is **linked list**.

Array

2	7	8	4	**Value**
0	1	2	3	**Index**

Linked lists

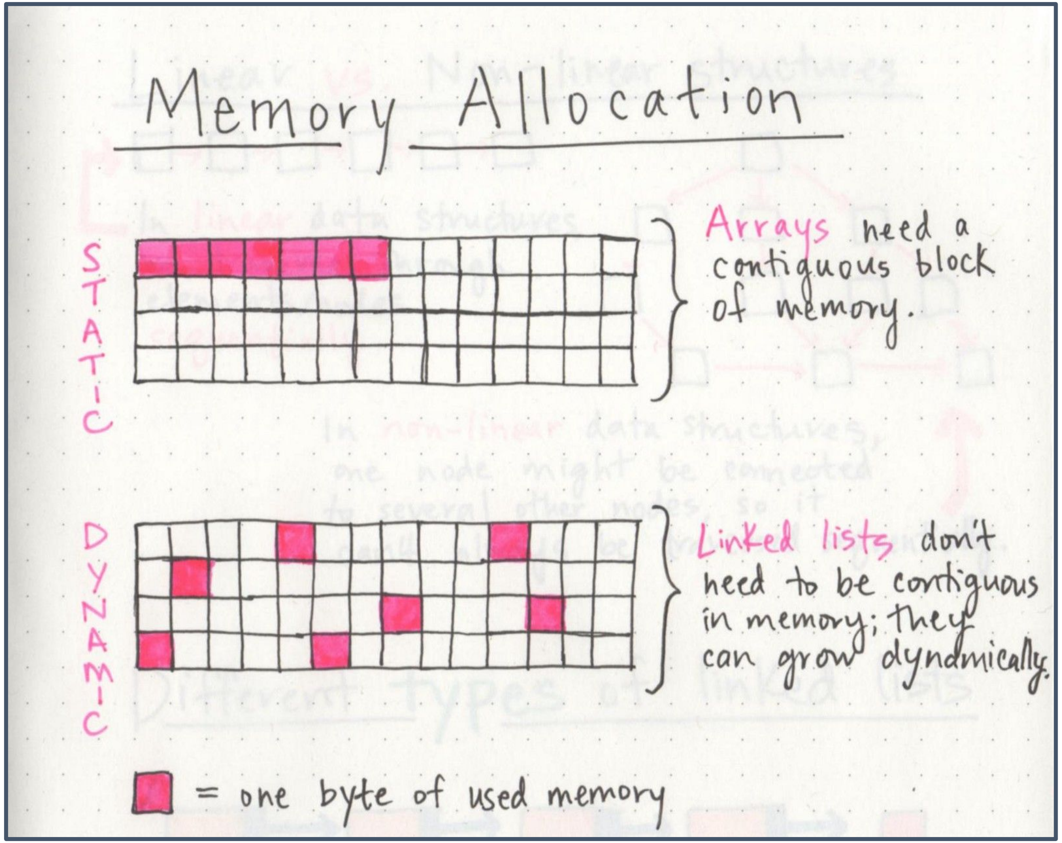
Linked Lists

Static

- Pre-allocated
- Fixed size
  - Unable to grow

Dynamic

- Allocated as needed
- Able to grow



Retrieved from https://sisyphus.gitbook.io/project/leetcode-notes/linked-list/difference-between-linked-list-and-array CC BY 2.0

Linked Lists

TYPE

- Singly Linked List
- Circularly Linked List
- Doubly Linked List
- Doubly Circularly Linked List

ACTION

- Insert
- Delete
- Search

What is a Linked List?

A singly **linked list** is a collection of nodes that form a linear order of a sequence [1].

[1] Michael T. Goodrich et al., Data Structures and Algorithms in Python, 2013

What is a Linked List?

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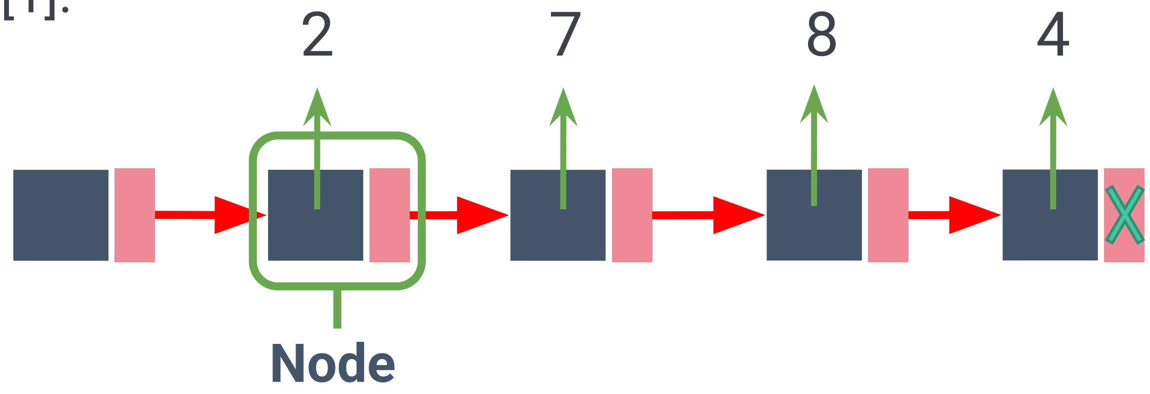
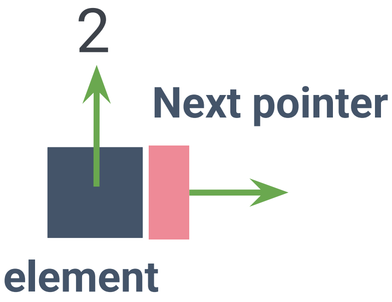


Linked List Node



Linked List Node Structures

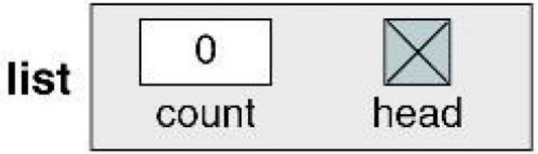
Create a Linked List

1. **Create a sentinel/header/root node**

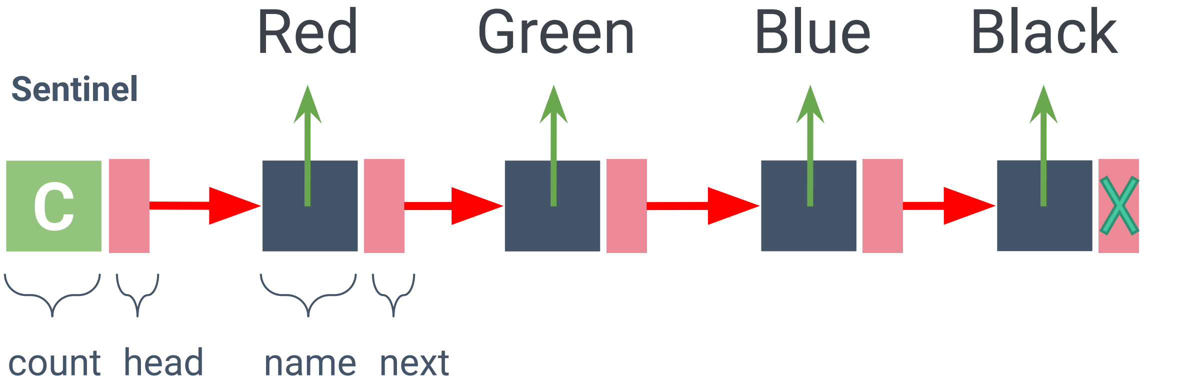
Algorithm createList (list)

1. Allocate a list
2. Set list head to null
3. Set list count to 0

End createList



Singly Linked Lists



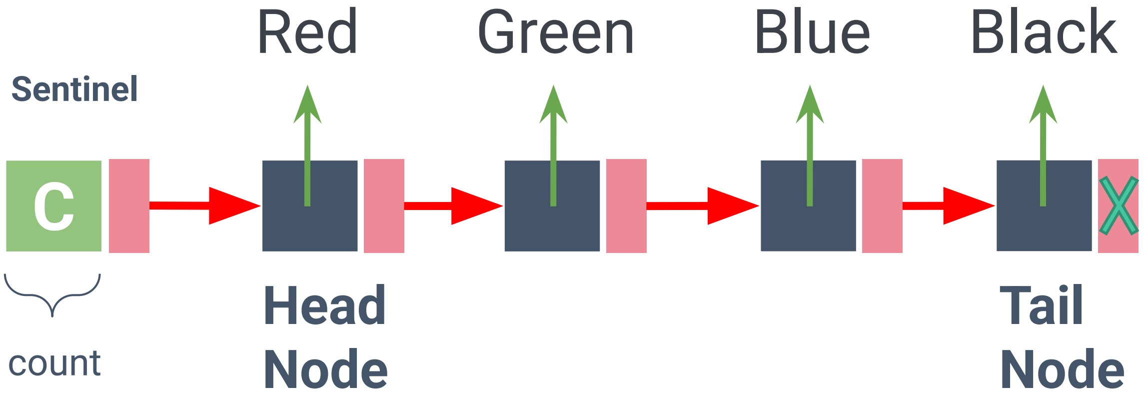
Color

String name

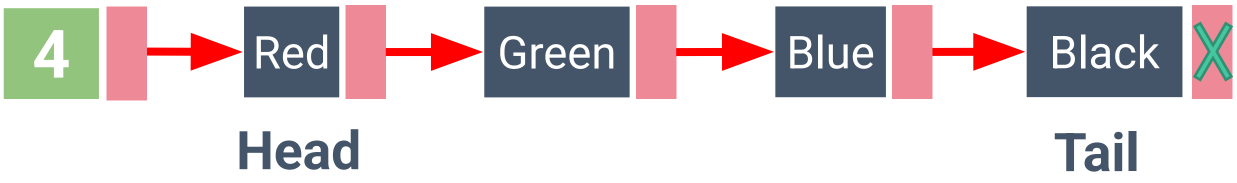
Color next

End Color

Singly Linked Lists

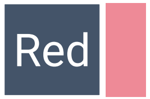


Singly Linked Lists



Create a Linked List

2. Create a data/element Node




```
Algorithm createDataNode (d, p)

    colorNew = allocate(Color)

    name = d

    next = p

    return colorNew

End createDataNode
```

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Traversing Singly Linked Lists

Address/ Byte#	Value
6000	4
6001	6002
6002	2
6003	6008
6004	8
6005	6012
6006	
6007	
6008	7
6009	6004
6010	
6011	
6012	4
6013	None

Suppose that it takes 1 byte to store an integer.

Traversing Singly Linked Lists

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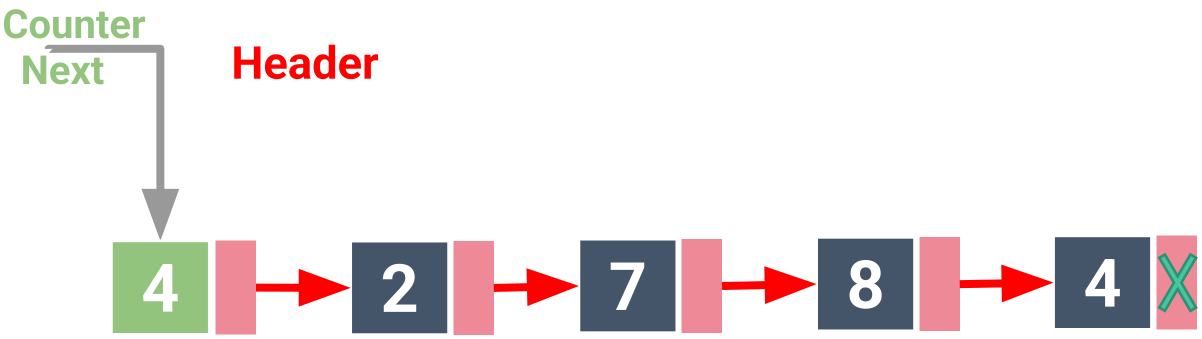

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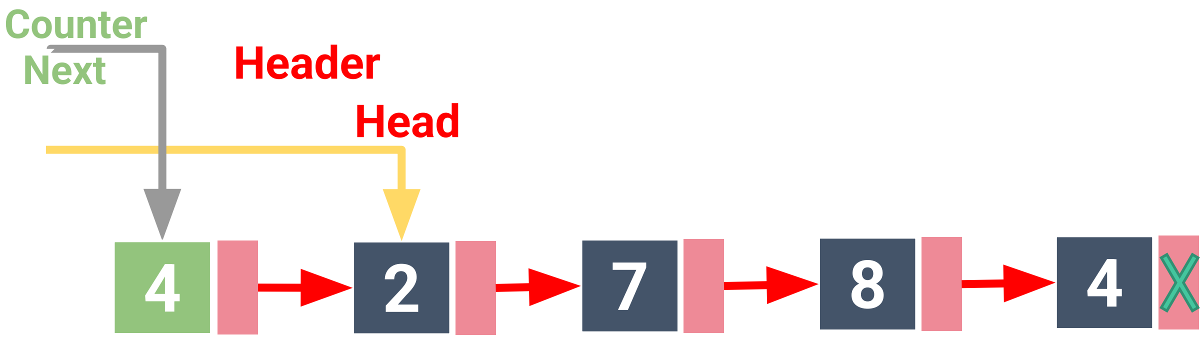

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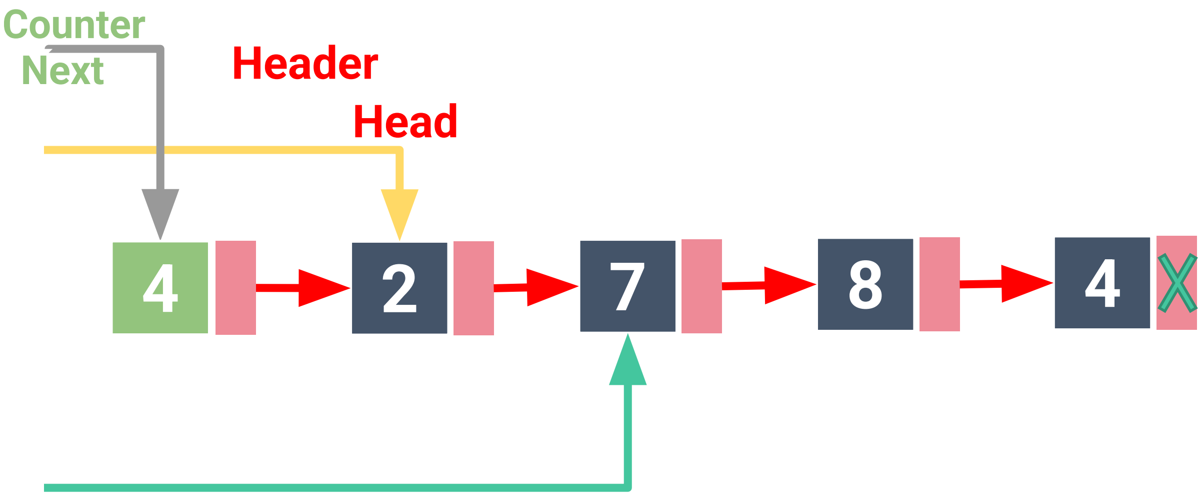

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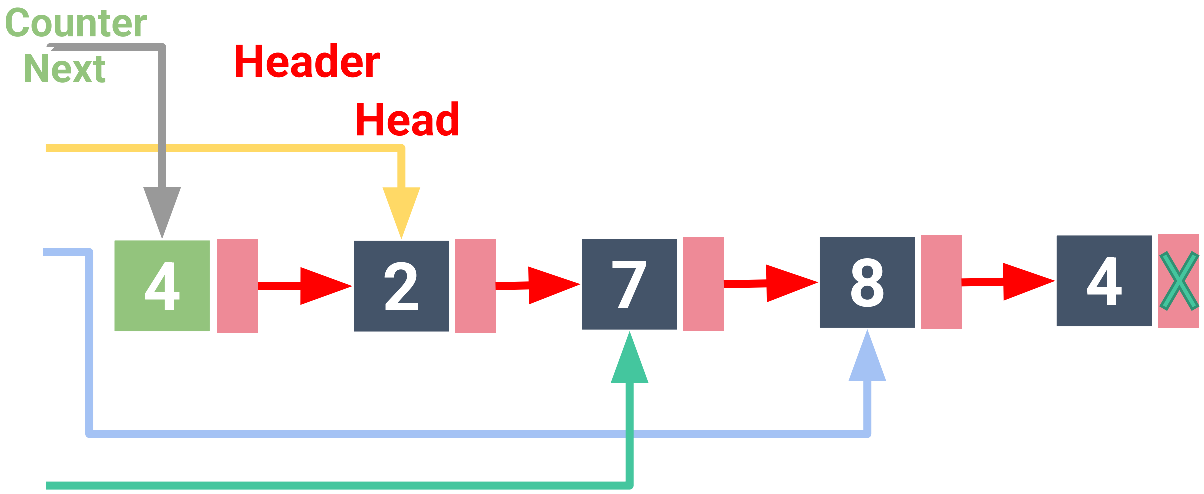

Traversing Singly Linked Lists

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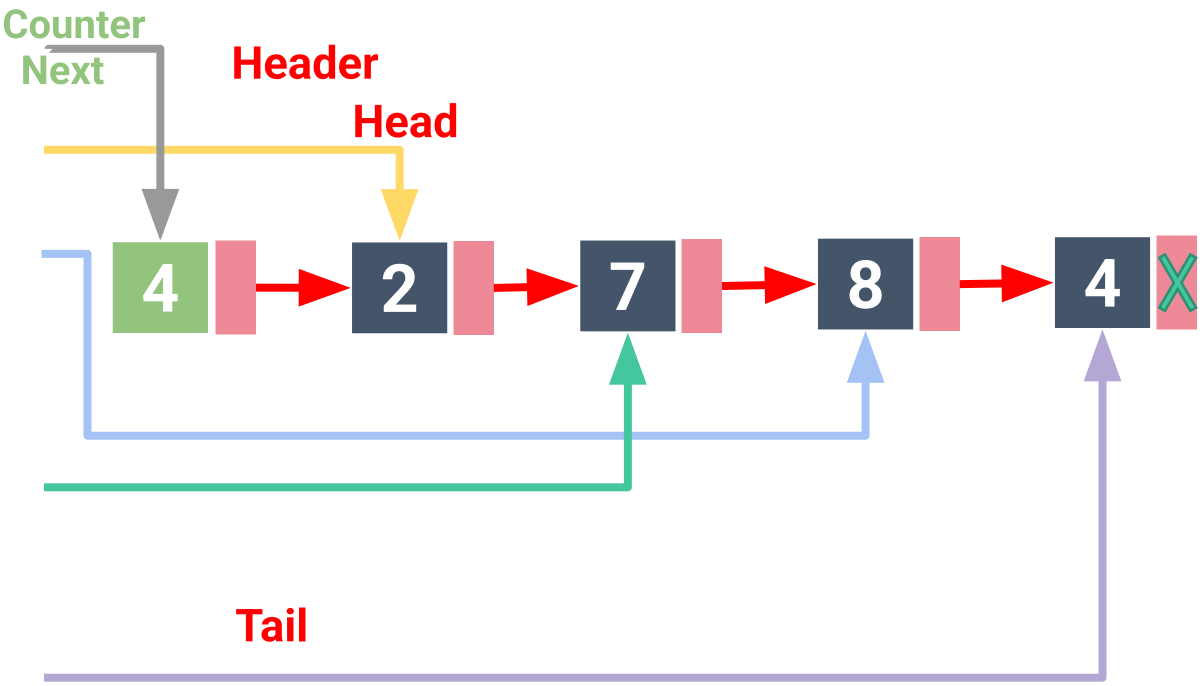

Linked List Operations

See the interactive guide on how insertion and deletion works